

Almost Binary Alms
2020 555 007

Dayline

1) $1 \leq a < 2$

$$x = 2^5 + 2^{10} + 2^{-19}$$

$$x = 2^5(1 + 2^{-15} + 2^{-24})$$

$$x = (1.0000 \dots 1 \dots 1)_2 \cdot 2^5$$

$$x = (1.000 \dots 1 \dots 1)_{a_5} \cdot 2^5 = 2^5(1 + 2^{-15})$$

$$x = (1.0000 \dots 1 \dots 1)_{a_5 \dots a_{23}} \cdot 2^5 = 2^5(1 + 2^{-15} + 2^{-23})$$

2.) $x = 0.643676895$
 $y = 0.643671734$

a) $x+y = ?$ x is digit

$$\left| \frac{(x+y) - [f_1(x) + f_1(y)]}{x+y} \right|$$

$$x^* = 0.643677 \quad y^* = 0.643672$$

$$x^* + y^* = 1.287349$$

$$100 \times \left| \frac{1.287348629 - 1.287349}{1.287348629} \right| = \text{Error}$$

$$= 0.00002881522213$$

$$= 0.000028$$

b) $x-y = ?$

$$\left| \frac{(x-y) - [f_1(x) - f_1(y)]}{x-y} \right|$$

$$\left| \frac{0.0000005161 - [0.0000005]}{0.000005161} \right| \times 100 = \text{Error}$$

$$\% 3.119550474714203$$

$$\% 3.1196$$

$$3.) f(x) = x^3 + 2x^2 - 5x - 6$$

$$x_0 = 1 \quad x_1 = -1$$

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

$$f(1) = -8 \quad f(-1) = 0$$

$$x_2 = -1 - 0 \cdot \frac{-1-1}{0-(-8)} = -1 + 0 = -1$$

$$4.) f(x) = x^3 + 2x^2 - 5x - 6$$

$$x_0 = 1$$

$$f'(x) = 3x^2 + 4x - 5$$

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1 - \frac{1+2-5-6}{3+4-5} = 1 - \frac{-8}{2} = 5$$

$$x_2 = 5 - \frac{5^3 + 2 \cdot 5^2 - 5 \cdot 5 - 6}{3 \cdot 5^2 + 4 \cdot 5 - 5}$$

$$x_2 = 5 - \frac{144}{20} = 5 - 7.2 = -2.2$$